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Numerical investigation of double-walled direct absorption evacuated tube solar collector using microencapsulated PCM and nanofluid

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ABSTRACT

In the present study, the thermal performance of direct absorption solar evacuated tube collectors utilizing CuO nanofluid, CuO/Al_2O_3 binary nanofluid, and CuO nanofluid combined with microencapsulated phase change material (MPCM) is investigated. The impact of base fluid types (water and ethylene glycol (EG)), nanofluid volume fraction, and flow velocity, are analyzed by the numerical simulation of the collector. Also, two models (single-phase and mixture) are used for simulating the MPCM slurry. Due to the insignificant difference between the results of the single-phase and two-phase models, the single-phase model is recommended to reduce computational time. It is found that the most significant differences in outlet temperature occur using 0.06 % Al_2O_3+ 0.002 % CuO/water nanofluid and 0.06 % Al_2O_3+ 0.002 % CuO/EG nanofluid which are equal to 28.34 K, and 36.64 K, respectively. The collector efficiency using binary nanofluids is higher than using CuO nanofluids in both water- and EG-based nanofluids. Results indicate that combining MPCM and CuO nanofluid increases the collector efficiency by 4.53 % and lowers the heat loss by 5.84 %. The simultaneous application of MPCM and CuO nanofluid has the highest efficiency among all working fluids.

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1. Introduction

As a result of industrialization and the world's rising population, the energy requirements of many civilizations have improved recently. In terms of energy harvesting and utilization, solar energy is an excellent renewable resource. Furthermore, solar energy has a lot of interest regarding its high ability to provide the energy needs of the world's population, at least in theory [1]. Solar collectors are the most significant components in converting solar energy into thermal energy. They work effectively as a heater by capturing solar energy and transferring it to the heat transfer fluid (HTF). Direct absorption solar collectors (DASC) have been recommended to improve solar collector efficiency. By using this collector, there is no need for an absorber plate. Nanoparticles distributed in the HTF absorb the incoming solar radiation directly. In terms of thermal efficiency, this type of collector is more effective than the conventional ones. Moreover, DASCs have reduced thermal resistance compared to indirect solar collectors [2–5].

DASCs based on nanofluids have undergone extensive research in order to improve their thermal performance. Guo et al. [6] conducted an experiment in which they inserted an optical fiber into a DASC to utilize as an internal light source in order to enhance thermal capabilities of water-based multiwall carbon nanotube (MWCNT) nanofluids. They observed that introducing optical fiber into the DASC enabled the MWCNT-H_2O nanofluids to have a consistent temperature distribution and improved the thermal performance of MWCNT-H_2O nanofluids. Moreover, at 0.010 wt% MWCNT and a 15 mm optical fiber location, an efficiency of 65.4 % was attained. Esmaeili et al. [7] numerically investigated water-saturated metal oxide foams and CuO nanofluid in the flat-plate DASC using the finite volume technique. They examined the impacts of volume fractions, mass flow rates, foam pore sizes and porous layer thicknesses and locations. According to the results, the best collector efficiency was attained when fully saturated with metal oxide. An Eulerian-Eulerian two-phase numerical simulation was used for the research conducted on the DASC by Balakin et al. [8]. As well as the concentration and size of nanoparticles, they examined collector geometry and inclination in a multi-parametric numerical analysis. The efficiency was raised by 10 % as a result of the optimization. According to their simulation results, the maximum efficiency of 64 % corresponded to φ_p = 10 ppm and d_p = 100 nm. The collector efficiency was also influenced by its position in the gravitational field. It reached the maximum value when the incident radiation was oriented in the opposite direction of the

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gravitational force. They also employed magnetic nanofluid, which boosted the collector performance by 30 %. In addition, Karami et al. [9] employed graphite nanofluid (GO) to explore the thermal performance of three types of collector which are conventional, forward- and backward-ribbed DASC respectively. They observed that a nanofluid with a volume fraction of 500 ppm was best for conventional DASC and forward-DASC in terms of energy and exergy efficiency. The conventional DASC, with its lower friction factor, had the highest exergy efficiency of the three collectors. Moreover, the exergy efficiency of DASC, backward-DASC, and forward-DASC for 500 ppm GO nanofluid was 7.6 %, 6.8 %, and 6.4 %, respectively. Siavashi et al. [10] executed another numerical analysis on the effectiveness of a flat-plate DASC employing an absorber plate and single-walled carbon nanohorn (SWCNH) utilizing a water-based fluid. They used the MRT-LBM code, which is a lattice Boltzmann approach, and discovered that the spectrum absorption of water was significantly improved when a low concentration of SWCNH nanoparticles was dispersed to the water. It was also found that employing an absorber plate for the collector instead of nanofluid positively influenced efficiency.

Flat-plate solar collectors (FPC) and evacuated tube solar collectors (ETC) are the most common collectors for residential solar applications. The weather and inlet temperature have a greater impact on the FPSC performance than on the ETC performance. Furthermore, one of the advantages of ETCs is their lower overall heat loss coefficient with respect to FPCs [11,12]. Sabiha et al. [13] and Kumar et al. [14] reviewed the advancements and most recent innovations in ETCs. They reported that an ETC outperforms FPC collectors in terms of efficiency. Therefore, the use of nanofluids and direct radiation absorption has the potential to improve collector efficiency significantly [15]. In the last decade, various researchers have evaluated the ability of nanofluids as HTFs in solar systems to enhance the efficiency of these systems. Tafarroj et al. [16] studied the computational fluid dynamics (CFD) and artificial neural network (ANN) model to evaluate the performance of a parabolic trough collector with direct absorption (DAPTC) using two different nanofluids (nanosilica and multi-walled carbon nanotubes). Findings revealed that the network and CFD results are in good agreement. Also, they concluded that the run time was reduced using an artificial network. O'Keeffe et al. [17] scrutinized the thermal efficiency of a DAPTC under a turbulent flow regime utilizing nanofluid. They discovered that the temperature at the outlet rose as the nanofluid passed through the absorber. Kasaeian et al. [18] used silica and carbon nanotube nanoparticles in EG to evaluate the thermal performance of the DAPTC. They investigated three different receivers and discovered that 0.3 % MWCNT/EG had the maximum efficiency in all tube absorbers types. Menbari et al. [19] studied the theoretical and experimental impacts of a CuO/water nanofluid on a DAPTC. According to the findings, increasing the nanofluid volume fraction and flow rate boosted efficiency. In the experimental work, done by Gupta et al. [20], the effect of flow rate on a DASC efficiency is studied using Al_2O_3/water nanofluid. Highest collector efficiency is obtained with a flow rate of 2.5 l/min for water and 2 l/min for 0.005 % volume fraction of Al_2O_3 nanoparticles. Moreover, Menbari et al. [21] performed the first experimental and theoretical studies on binary nanofluid, which was a combination of CuO and Al_2O_3 nanofluids. As demonstrated by the results, the absorption coefficient of binary nanofluid is almost identical to the total absorption coefficients of its components. In another numerical study, Kasaeian et al. [22] used MWCNT nanoparticles to scrutinize the effectiveness of a parabolic trough collector (PTC). Findings showed that a 0.5 % volume fraction of nanoparticles caused an 8.6 % enhancement in thermal efficiency and a 6.5 K improvement in outlet temperature difference when DASC was used instead of the conventional PTC.

Recently, there has been a lot of interest in phase change materials (PCMs) due to their great potential for storing thermal energy. During the melting process of the PCMs, the thermal energy is stored at a constant temperature throughout the transition (There is no change in sensible heat) [23,24]. A number of papers have discussed the utilization of PCMs in solar energy applications. Hasan et al. [25] investigated the annual energy savings potential of a photovoltaic-phase change material (PV-PCM) panel experimentally. They discovered that PCM lowered the temperature of PV panels by 10.5 °C. In addition, electrical energy grew by 5.9 % per year. Feliński and Sekret [26] presented the experimental research using an evacuated tube collector (ETC) to investigate the effect of paraffin. They reported that paraffin reduced the working fluid temperature, significantly affecting the solar collector heat loss. Additionally, when paraffin incorporated into the ETC system, the total generated heat is increased by 45–79 % compared to the conventional ETC.

Some other studies in the literature suggested using MPCM slurry, which is a dispersion of MPCM in the fluids, to promote the performance of solar collectors. The MPCM is made by encasing organic PCMs as the core within an organic or inorganic shell, and they can be inserted into the based fluid. The microencapsulation method of the PCM avoids leakage all over the phase change transition and improves PCM thermal conductivity [27]. It is possible to use slurry as both a heat transport fluid and a thermal energy storage medium. The high heat capacity of MPCM slurry has boosted the system efficiency. Liu et al. [28] conducted another numerical research on a novel PVT collector utilizing MPCM slurry. At 11:00 a.m., the MPCM slurry collector overall efficiency was 80.57 %, which was around 1.8 % greater than the efficiency of traditional water-based collectors. A numerical investigation of the FPSC was conducted by Serale et al. [29] using a single-phase model of the MPCM slurry. They found that by utilizing MPCM slurry as a HTF, they were able to increase solar energy utilization. Additionally, the collector efficiency was boosted by adding larger concentration of the MPCM slurry. Qiu et al. [30] theoretically explored a PV/thermal (PVT) system efficiency by utilizing MPCM slurry. Under turbulent flow conditions, it was reported that the temperature of the PV module was dropped, and the system overall efficiency raised by improving the slurry concentration ratio. Furthermore, at a 5 % concentration ratio, the PVT module efficiency was the highest.

Recently, some experimental works have been conducted on the application of MPCM slurry to the DASC. Zhang et al. [31] modified the MPCM shell with Ti_4O_7 nanoparticles. Based on findings, the microencapsulated' photo-thermal storage efficiency was 85.36 %. Wang et al. [32] evaluated the thermal ability of a paraffin/water phase change emulsion by incorporating graphite nanoparticles. They found that 0.1 % wt graphite increased its thermal conductivity by 20.0 % with regard to the base fluid. Conclusively, the emulsion of paraffin/water/graphite exhibited high photo-thermal performance.

Several research works have investigated the flat-plate DASC based on the literature review on enhancing the collector thermal performance. However, there are only a few articles about the MPCM slurry-based direct absorption evacuated tube solar collector (DAETC). Therefore, in this paper, MPCM-based DAETC is simulated using ANSYS-Fluent 2021 R2. Besides, the effect of several parameters, such as nanofluid types, different base fluids, volume fractions, and flow velocity is investigated. The CuO nanofluid with two volume fractions (0.001 % and 0.002 %) and the CuO/Al_2O_3 binary nanofluid, in different base fluids are selected as the working fluids. In the previous investigations, the thermal behavior of MPCM is reported using the single-phase model. However, in this paper, the two-phase model is also used to simulate the MPCM,

and the thermal performance of the collector is compared using the two models.

2. System description

A three-dimensional model of the DAETC is developed to assess the collector thermal performance in the present work. Fig. 1 depicts the schematic of the DAETC. The detailed dimensions of the proposed collector are presented in Table 1. Borosilicate glass is utilized for the absorber tube, of which the thermophysical properties are shown in Table 2. It is assumed that solar energy enters the collector at 90° to the horizon. The HTF flows in the internal section, and the vacuum section plays the role of thermal insulation. Water and ethylene glycol are the most common base working fluids utilized in the DASCs. Ethylene glycol is added to water to avoid freezing water in cold weather. Due to the low absorption coefficient of these base fluids, inserting nanoparticles into them improves both the optical characteristics and the thermal efficiency of the collector [33]. The thermophysical properties of the mentioned base fluids are listed in Table 2. As shown in Table 3, five different HTF including CuO/water nanofluid, CuO/EG nanofluid, binary (CuO + Al_2O_3)/water nanofluid, binary (CuO + Al_2O_3)/EG nanofluid in two different volume fractions, and CuO/water nanofluid with MPCM are investigated. It is worth noting that the extinction (absorption + scattering) coefficient of nanofluids is extracted from the experimental work, done by Menbari et al. [21,34] and shown in Table 4.

In the present work, the PCM is hydrocarbon n-eicosane and the shell material of the MPCM is TiO_2. Table 5 shows the characteristics of the MPCM. The effective thermophysical properties of the MPCM slurry are calculated utilizing the following equations [35]:

$$\rho_{eff} = \rho_{pcm}\alpha_p + (1-\alpha_p)\rho_{bf} \tag{1}$$

$$\mu_{eff} = \mu_f\left(1-\alpha_p-1.16{\alpha_p}^2\right)^{-2.5} \tag{2}$$

$$k_{eff} = \left[\frac{k_{pcm}+2k_{bf}+2\left(k_{pcm}-k_{bf}\right)\alpha_p}{k_{pcm}+2k_{bf}-2\left(k_{pcm}-k_{bf}\right)\alpha_p}\right]\times k_{bf} \tag{3}$$

$$c_{p.eff} = \frac{\alpha_p\rho_{pcm}}{\rho_{pcm}\alpha_p+(1-\alpha_p)\rho_{bf}}c_{p.pcm} + \frac{(1-\alpha_p)\rho_f}{\rho_{pcm}\alpha_p+(1-\alpha_p)\rho_{bf}}c_{p.bf} \tag{4}$$

where α_p, ρ_{eff}, μ_{eff}, k_{eff} and $c_{p.eff}$ are solid volume fraction, the effective density, viscosity, thermal conductivity, and specific heat capacity of the MPCM slurry. Also, the PCM particle and working fluid are denoted by the subscripts "p" and "f," respectively.

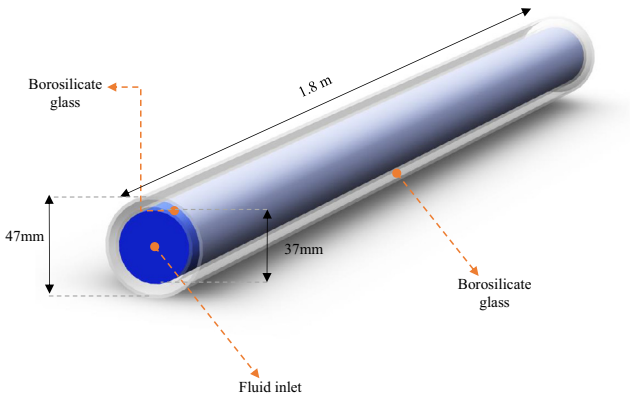


Fig. 1. Three dimensional model of DAETC.

Table 1
Dimensions of DAETC.

Parameter	Dimension
Length	1.8 m
Inner diameter of internal tube	33.8 mm
Outer diameter of internal tube	37 mm
Inner diameter of external tube	43.8 mm
Outer diameter of external tube	47 mm

Table 2
Thermophysical properties of collector materials.

Material	Density ($\frac{kg}{m^3}$)	Thermal conductivity ($\frac{W}{m.K}$ $\frac{W}{m.K}$)	Specific heat ($\frac{J}{kg.K}$ $\frac{W}{m.K}$)
Borosilicate glass	2124.9	1.1489	779.74
Water	998.2	0.6	4182
Ethylene glycol	1111	0.253	2415

Table 3
Different working fluids used in this study.

Working fluid	Label
0.001 % CuO/water nanofluid	C1
0.002 % CuO/water nanofluid	C2
0.001 % CuO/ethylene glycol nanofluid	C3
0.002 % CuO/ethylene glycol	C4
Binary (0.04 % Al_2O_3 + 0.001 % CuO)/water	B1
Binary (0.06 % Al_2O_3 + 0.002 % CuO)/water	B2
Binary (0.04 % Al_2O_3 + 0.001 % CuO)/ethylene glycol	B3
Binary (0.06 % Al_2O_3 + 0.002 % CuO)/ethylene glycol	B4
MPCM + 0.002 % CuO/water	M1

During the solid-to-liquid transition, the effective specific heat capacity of MPCM is dependent mostly on its latent heat. For evaluating the effective specific heat capacity of the MPCM, multiple models have existed. In this study, the middle triangle model is adopted to compute this property, as shown in Fig. 2. Also, the following equations are given in this model [35]:

$$\int_{Tpcm.s}^{Tpcm.l} c_{p_{pcm}}dT = \Delta H_{pcm} + \frac{c_{p.pcm.s}+c_{p.pcm.l}}{2}\left(T_{pcm.l}-T_{pcm.s}\right) \tag{5}$$

$$c_{p_{pcm}} = \left(T_{pcm}-T_{pcm.s}\right)\left[\frac{c_{p.pcm.l}-c_{p.pcm.s}}{T_{pcm.l}-T_{pcm.s}} + \frac{4\Delta H}{\left(T_{pcm.l}-T_{pcm.s}\right)^2}\right] + c_{p.pcm.s}$$
$$\rightarrow T_{pcm.s} \le T_p \le \frac{T_{pcm.s}+T_{pcm.l}}{2} \tag{6}$$

$$c_{p_{pcm}} = \left(T_{pcm.l}-T_{pcm}\right)\left[\frac{4\Delta H}{\left(T_{pcm.l}-T_{p.m1}\right)^2} - \frac{c_{p.pcm.l}-c_{p.pcm.s}}{T_{pcm.l}-T_{pcm.s}}\right] + c_{p.pcm.l} \rightarrow \frac{T_{pcm.s}+T_{pcm.l}}{2} \le T_p \le T_{pcm.l} \tag{7}$$

where ΔH is the latent heat, and the subscript "s" and "l" denote the solid and liquid states of the MPCM, respectively. According to the MPCM, the lower limit is defined $T_{pcm.s} = T_m - \Delta T$, while the higher limit is defined as $T_{pcm.l} = T_m + \Delta T$ [35]. Note that ΔT is equal to 1 K for the MPCM slurry.

Table 4
Extinction coefficient of nanofluids.

Nanofluids	C1	C2	C3	C4	B1	B2	B3	B4
Mean extinction coefficient ($\frac{1}{m}$)	165	419	139	341	288	516	233	397

Table 5
Thermophysical properties of MPCM [36].

Density ($\frac{kg}{m^3}$)	Specific heat ($\frac{J}{kg.K}$)	$\frac{W}{m.K}$ Melting point (K)	Thermal conductivity ($\frac{W}{m.K}$)	Latent heat of fusion (J/kg)	Particle diameter (μm)
946.4	1973.1	310	0.749	192,660	10

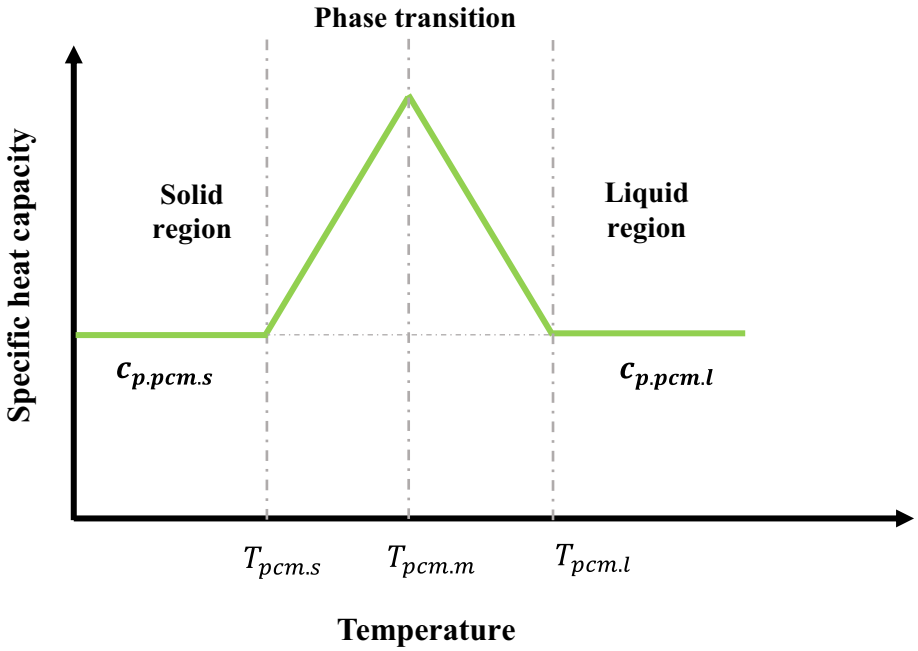


Fig. 2. Temperature-dependent variation of MPCM specific heat capacity [37].

3. Governing equations and numerical simulation

Fluid flow is supposed to have a steady, laminar, and incompressible flow regime. The numerical simulation is made more straightforward by considering the following assumptions:

- The nanofluid is considered to have the same thermophysical properties as the base fluid.
- It is presumed that the nanoparticles will remain suspended in the fluid [3].
- The impact of gravity is ignored.

To simulate the MPCM slurry in the DAETC, the single-phase model is commonly applied to simplify the numerical simulation. Therefore, the MPCM is considered a single-phase fluid [35,36,38]. Also, MPCM with less than 25 % concentration can be assumed as a Newtonian fluid [39,40]. The single-phase model is utilized based on the value of the thermophysical properties such as density, thermal conductivity, viscosity, and specific heat capacity of the MPCM. Besides the single-phase model, the mixture two-phase model is utilized to compare the heat transfer behavior of the MPCM. There are significant differences in the equations of continuity, momentum, and energy conservation between the single-phase and the mixture models. The following formulations describe the governing equations of both models:

(a) Single-phase model

The following are the continuity, momentum, and energy conservation equations for the fluid domain [41]:

$$\nabla \cdot \left(\overrightarrow{V_f} \right) = 0 \tag{8}$$

$$\rho_f \left[(\overrightarrow{V_f} \cdot \nabla) \overrightarrow{V_f} \right] = -\nabla P + \nabla \cdot (\mu_f \nabla \overrightarrow{V_f}) \tag{9}$$

$$\rho_f C_{p,f} \overrightarrow{V_f} \cdot \nabla T_f = \nabla \cdot (k_f \nabla T_f) + q_s \tag{10}$$

where $\overrightarrow{V_f}$, P, μ,k and q_s are fluid velocity, pressure, viscosity, thermal conductivity and volumetric energy source term, respectively.

(b) Mixture model

In terms of complexity, the two-phase model has a higher level of sophistication than the single-phase model because the MPCM is considered as a separate phase. The mixture model continuity equation is found as:

$$\nabla \cdot \left(\overrightarrow{v_m} \right) = 0 \tag{11}$$

$$\overrightarrow{v_m} = \frac{\sum \alpha_k \rho_k \overrightarrow{v_k}}{\rho_m} \tag{12}$$

$$\rho_m = \sum_{k=1}^{n} \alpha_k \rho_k \tag{13}$$

where $\overrightarrow{v_m}$ is the mass-averaged velocity, ρ_m is the mixture density and α_k is the volume fraction of the phase k.

Summing the momentum equations for each phase provides the mixture model's momentum equation, which is given below [41]:

$$\left[\nabla \cdot (\rho_m \overrightarrow{v_m} \overrightarrow{v_m})\right] = -\nabla P + \nabla \cdot [\mu_m \left(\nabla \overrightarrow{v_m} + \nabla \overrightarrow{v}_m^T\right] + \rho_m \overrightarrow{g} + \overrightarrow{F} + \nabla \cdot \left(\sum_{k=1}^{n} \alpha_k \rho_k \overrightarrow{v_{dr.k}} \overrightarrow{v_{dr.k}}\right) \tag{14}$$

$$\mu_m = \sum_{k=1}^{n} \alpha_k \rho_k \tag{15}$$

$$\overrightarrow{v_{dr.k}} = \overrightarrow{v}_k - \overrightarrow{v}_m \tag{16}$$

where n, $\overrightarrow{F}$ and μ_m represent the number of phases, body force, and the mixture viscosity. It should be noted that the slip velocity is not considered in this study. Following is a formulation that can be made for the energy equation that represents the mixture model [41]:

$$\nabla \cdot \sum_{k=1}^{n} \left(\alpha_k \overrightarrow{v_k} (\rho_k E_k + p)\right) = \nabla \cdot \left[k_{eff} \nabla T\right] + q_s \tag{17}$$

$$E_k = h_k - \frac{p}{\rho k} + \frac{v_k^2}{2} \tag{18}$$

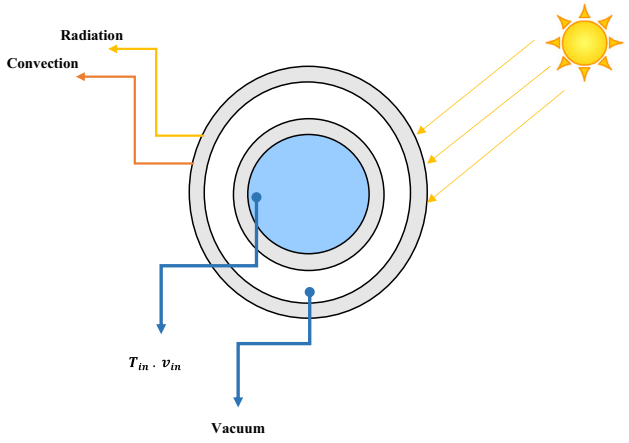


Fig. 3. Boundary conditions for modeling DAETC.

where k_{eff} is the effective thermal conductivity. The volumetric heat sources denoted by q_s and the sensible enthalpy for phase k represented by h_k, respectively. If the flow is incompressible, Eq. (18) is simplified to $E_k = h_k$ [41].

The energy source term is computed by solving the radiative transfer equation (RTE). To simplify the radiative transfer equation, the emission and scattering coefficient are neglected. By using the Beer-Lambert law, the volumetric energy source term takes the following form [8]:

$$q_s = I_0 K_{e\lambda} \exp(-(K_{e\lambda})l) \tag{19}$$

where I_0 denotes the solar irradiation, which corresponds to a value of 1000 $\frac{W}{m2}$.$K_{e\lambda}$ and l represent the extinction coefficient of the nanofluid and light path length in the collector, respectively. Moreover, the corresponding boundary conditions can be written as follows:

$$T = T_{in} \ . \ v = v_{in} \tag{20}$$

$$P_{out} = 0 \tag{21}$$

$$u = v = 0 \, (No \, slip) \tag{22}$$

where T_{in} and v_{in} are the inlet temperature and velocity of the HTF, which are assumed 293 K and 0.001 $\frac{m}{s}$, respectively. The ambient and sky temperatures are assumed 288.15 K and 270 K, respectively. The "inlet velocity" and "inlet temperature" are taken into consideration for the fluid moving inside the inner tube. The "pressure outlet" is also used in the collector outlet section. Besides, the tube wall is subjected to the no-slip condition. The upper and lower surface of the collector is considered as a "wall". An expression method is used to compute the energy source term and incorporate it into ANSYS-Fluent. The boundary conditions are depicted in Fig. 3.

Heat loss of the collector is an inevitable process. Three different mechanisms, conduction, convection, and radiation, are capable of transferring heat. Heat transfer methods must be considered in order to perform heat balancing. Heat loss, Q_{loss}, can be written as:

$$Q_{loss} = S - Q_{useful} \tag{23}$$

where Q_{useful} is [13]:

$$Q_{useful} = \dot{m} c_p (\bar{T}_{out} - T_{in}) \tag{24}$$

The heat lost between the collector and ambient air are obtained using the following relation [42]:

$$Q_{loss} = U_L \times A \times (T_{mean} - T_{amb}) \tag{25}$$

where T_{mean} is the average collector temperature, T_{amb} is the ambient temperature, A is the collector surface area, and U_L is the overall heat transfer coefficient.

In this paper, the collector is divided into two segments (upper surface and lower surface). An electrical analogy model (Fig. 4) is used for both sections in order to calculate the overall heat transfer

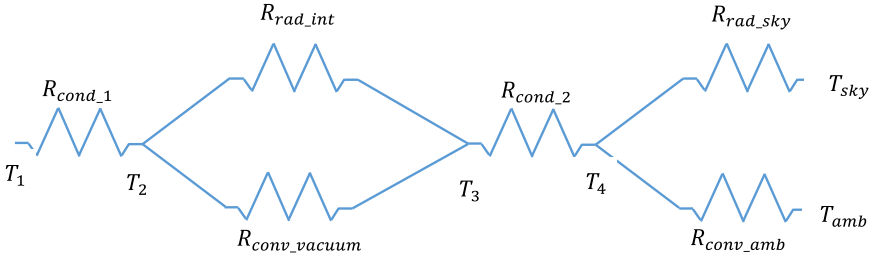


Fig. 4. Thermal network model of collector.

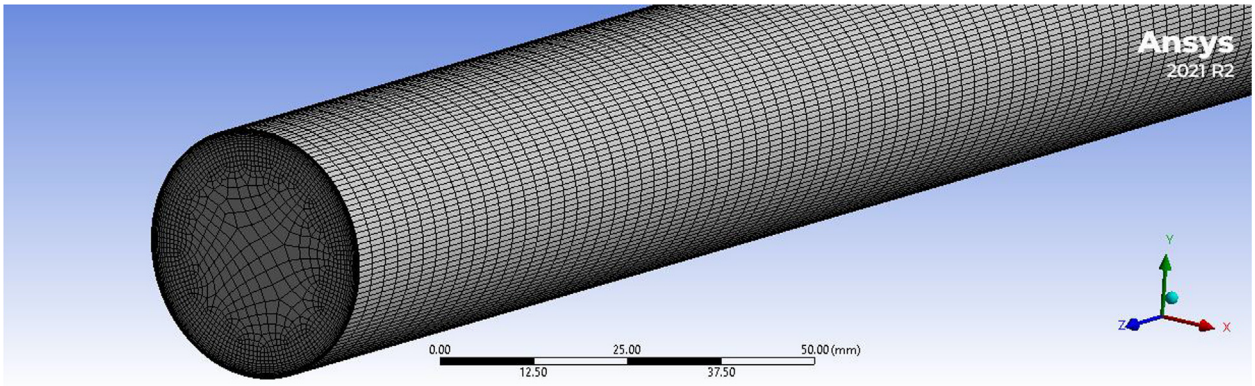


Fig. 5. Grid distribution of the collector.

coefficient [43,44]. Also, it is considered that the temperatures of the upper and lower surface of the collector are different ($T_{lower} = T_{amb}$, $T_{upper} = T_{sky}$).

The following equation can be utilized to compute the collector efficiency:

$$\eta = \frac{\dot{m}c_p(\bar{T}_{out} - T_{in})}{I_0 A} \tag{27}$$

where $\dot{m}$, c_p, $\bar{T}_{out}$, T_{in},I_0 and A is the mass flow rate, specific heat capacity, average outlet temperature, inlet temperature, considered irradiance and collector net surface, respectively.

A numerical simulation is carried out using ANSYS-Fluent 2021 R2, which uses the finite volume method to discretize the continuity, momentum, and energy equations. The pressure base technique is chosen to solve the governing equations. The second-order upwind technique is employed to discretize the pressure and momentum equations. The pressure and velocity equations are coupled using the SIMPLE method [41]. The residuals of the continuity and momentum equations have convergence criteria of 10^{-5}, while the energy equation has convergence criteria of 10^{-8}.

ANSYS-workbench 2021 R2 is adopted to generate the grids for the numerical simulation. The result is shown in Fig. 5. Different grid numbers are performed to examine the grid independency of the findings. The three-dimensional structured mesh created for the computational domain with 300,000, 600,000, 1,300,000, and 1,600,000 elements is investigated. Also, the boundary layer mesh close to the collector inner surface is applied to improve

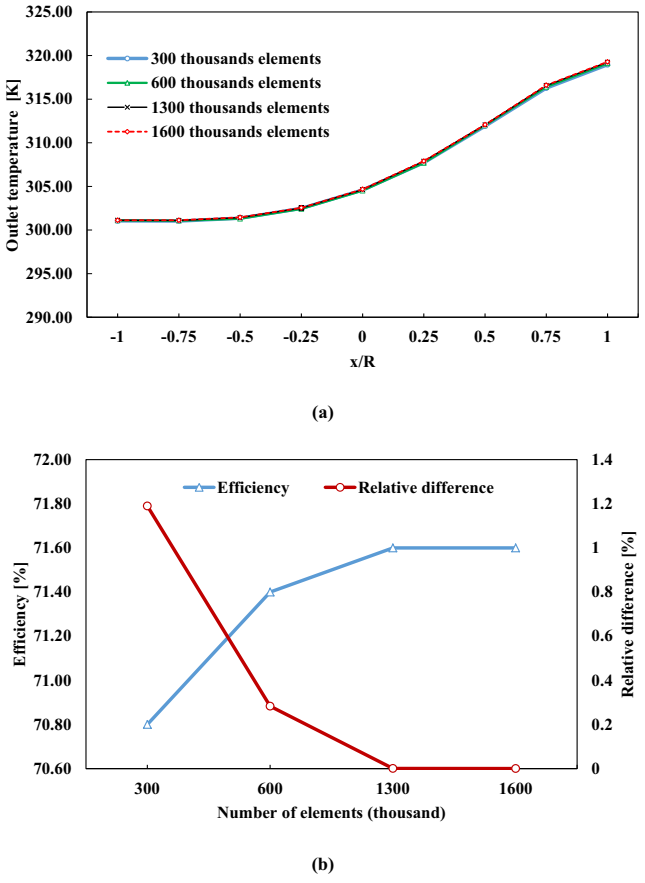


Fig. 6. Grid independence analysis for (a) outlet temperature, (b) collector efficiency and its relative difference.

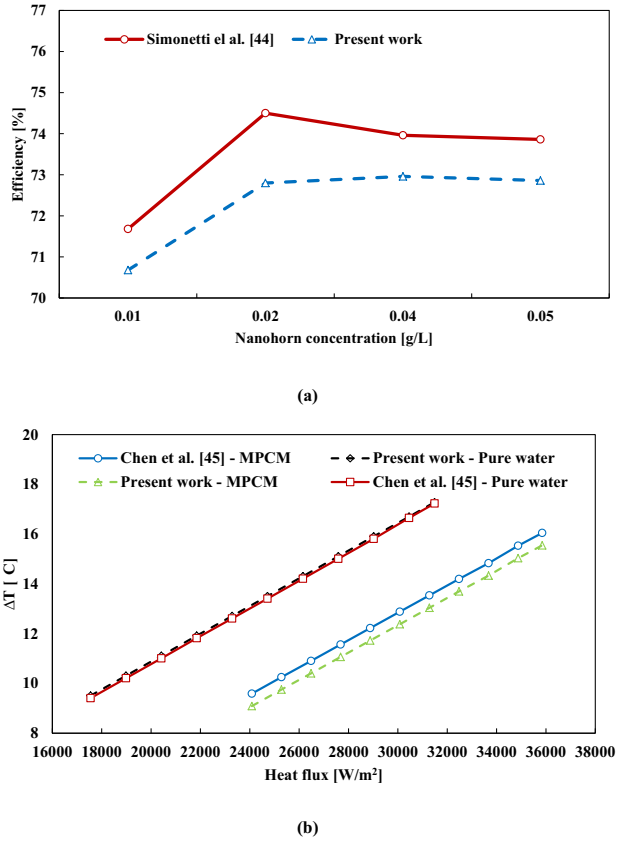


Fig. 7. (a) Comparison of the DAETC efficiency with the results of Simonetti et al. [44] and (b) comparison of the variation of fluid temperature rise with the results of Chen et al. [45].

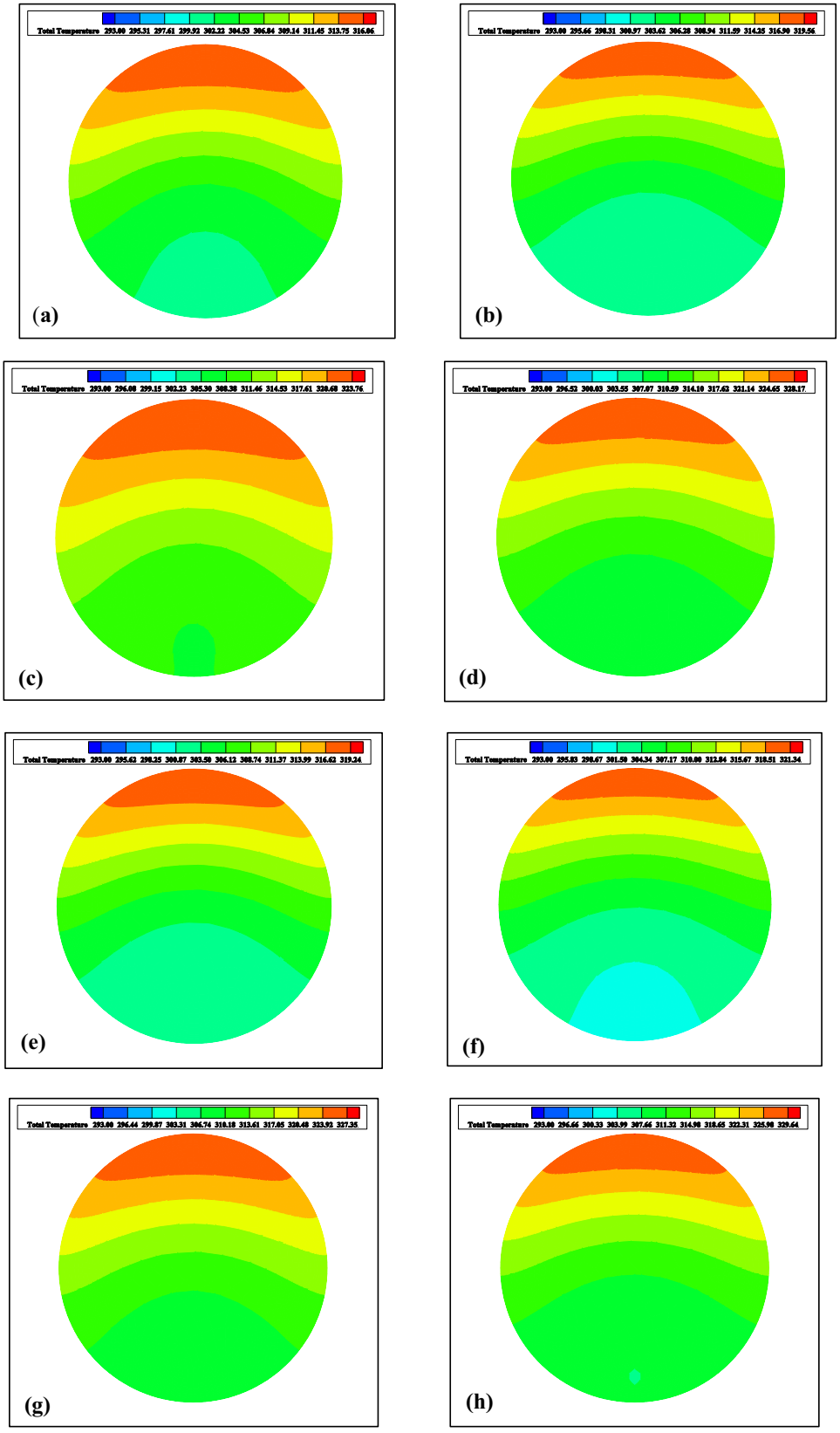


Fig. 8. Temperature contours at the collector outlet using different working fluids (a) C1, (b) C2, (c) C3, (d) C4, (e) B1, (f) B2, (g) B3, and (h) B4.

the accuracy of the simulation. Fig. 6 (a) presents the grid independence analysis at the outlet temperature profile of the collector using binary nanofluid (B1). Also, Fig. 6 (b) shows the collector efficiency and its relative difference by utilizing 1,600,000 elements as a reference value. As observed, 1,300,000 elements are suitable for further calculations due to negligible difference after this number

of mesh. The efficiency of the collector using 1,300,000 elements is so close to 1,600,000 elements; therefore, the former is considered to reduce the computational time of the simulation.

To ensure the accuracy of the simulation findings, a comparison of the DAETC performance with that of Simonetti et al. [44] is carried out. They examined the effect of the single-walled carbon nanohorn (SWCNH) concentration in the EG-based fluid on the collector efficiency. Based on Fig. 7 (a), the percentage error is less than 3 % which presents the acceptable accuracy of the present model.

Furthermore, the experimental study, done by Chen et al. [45], is utilized to confirm that the outcomes of the MPCM slurry simulation are valid. The ability of MPCM slurry to flow through a circular tube made of stainless steel while being subjected to constant heat flux was examined by Chen et al. [45]. The circular stainless steel tube (Cri8Ni9Ti) is 1460 mm in length, 4 mm in diameter, and 1 mm thick. As a PCM, 15.8 % of 1-bromohexadecane (C16H33Br) is selected, with an 8.2μ m particle diameter. Fig. 7 (b) presents the variation of the fluid temperature inside the circular tube at different heat fluxes and mean velocity of 0.64 $\frac{m}{s}$. As observed in Fig. 7 (b), for MPCM slurry modeling, the results of this study agree well with those of Chen et al. [45].

4. Results and discussion

In this section, the impact of several nanofluid types with different base fluids, volume fractions, and flow velocity on the collector performance is evaluated. In addition, the efficacy of MPCM on the collector thermal performance is analyzed and compared using both single-phase and two-phase models. The collector efficiency using binary nanofluid, which is the mixture of two nanofluids, and a combination of MPCM and nanofluid is also studied and compared.

4.1. Effect of working fluids

Fig. 8 presents the temperature contour at the collector outlet using C1, C2, C3, C4, B1, B2, B3, and B4 nanofluids, respectively. The HTF temperature has the lowest value at the collector inlet. The base fluid contains nanoparticles that absorb solar energy and boost the temperature of the working fluids. Based on Fig. 8 (a), the inlet temperature is 293 K, and the temperature is increased by 23.06 K ($\Delta T = T_{max} - T_{in}$). The increased volume fraction of CuO nanofluid from 0.001 % to 0.002 % improves the temperature differential at the outlet from 23.06 K to 26.56 K, as shown in Fig. 8 (a) and (b). This is because the nanofluid extinction coefficient is enhanced by increasing the volume fraction. Consequently, a more significant amount of heat is absorbed by the nanoparticles, which increases the temperature of the HTF.

Fig. 8 (c) depicts the outlet temperature contour of the collector using C3 nanofluid, which is an EG-based nanofluid. By comparing the temperature contour of the same volume fraction for CuO nanofluid in the water-based nanofluid (C1), it can be inferred that the outlet temperature in the water-based nanofluid is lower than the EG-based one. The outlet temperature difference for C1 is 23.06 K, while C3 is 30.76 K. The temperature contour at the collector outlet using C4 is shown in Fig. 8 (d). In this case, the outlet temperature difference is 35.17 K, while for C2, is 26.56 K.

By using the binary nanofluids (B1 and B2), the maximum outlet temperature is increased by 1 % and 0.55 % compared to that using C1 and C2, respectively (Fig. 8 (e) and (f)). The maximum outlet temperature increment using B3 and B4 nanofluids compared to that using C3 and C4 is 1 % and 0.44 %, respectively (Fig. 8 (g) and (h)). It is worth noting that as the nanofluid volume fraction increases, the heat dissipation improves, and the average fluid temperature at the outlet rises. The amount of energy received by nanofluid reduces exponentially with the optical path length. Due to this phenomenon, the working fluid temperature on the surface is significantly greater than at the collector's bottom for all types of nanofluids.

Fig. 9 reports the temperature characteristic of the collector using different HTFs. According to this figure, applying a small amount of CuO nanofluid volume fraction significantly affects the collector outlet temperature. The average temperature difference between binary/water nanofluids (B1 and B2) is 0.02 K. It is possible to conclude that there is no discernible rise in temperature associated with an increment in binary nanofluid volume fraction. Although, the temperature enhancement at the low volume fraction of CuO nanofluids grows more. This phenomenon also happened to EG-based nanofluids, as shown in Fig. 9. This indicates that raising the

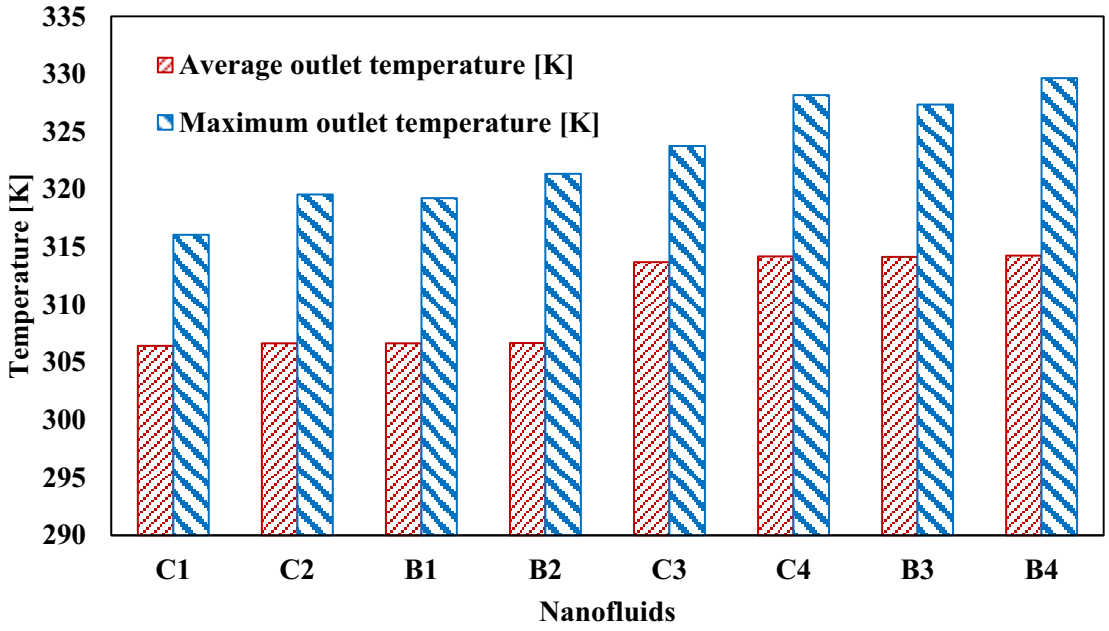


Fig. 9. Temperature characteristics of the collector using various working fluids.

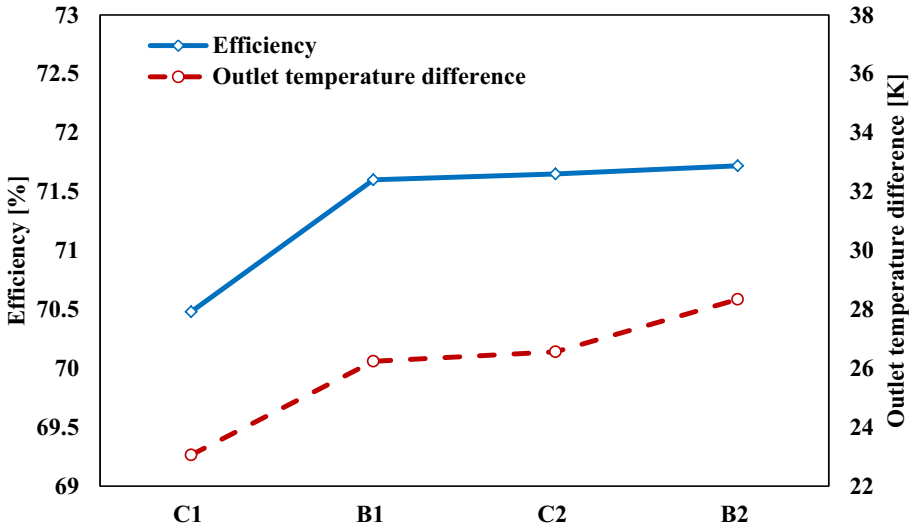


Fig. 10. Effect of nanoparticles volume fractions on collector efficiency and outlet temperature difference using water-based nanofluids.

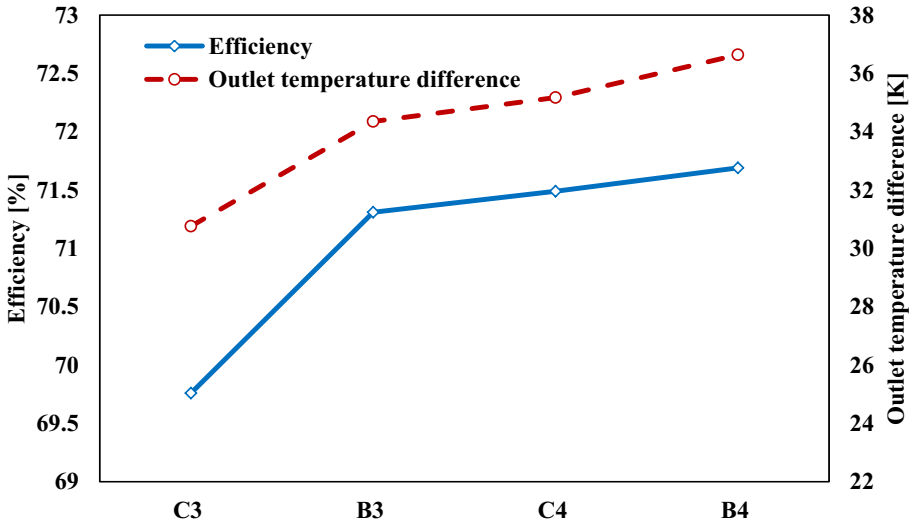


Fig. 11. Effect of nanoparticles volume fractions on the collector efficiency and outlet temperature difference using EG-based nanofluids.

volume fraction does not permanently improve the temperature at the collector outlet and requires an optimal value.

Fig. 10 illustrates the efficiency and the outlet temperature difference of the DAETC for mentioned water-based nanofluids. The collector efficiency using C1 and C2 is 70.48 % and 71.65 %, respectively. Also, the efficiency of the collector using B1 and B2 is 71.60 % and 71.72 %, respectively. According to Fig. 10, enhancing the nanofluid volume fraction causes an increment in the collector efficiency. This is due to the increase in the nanofluid extinction coefficient, which causes a greater temperature at the collector outlet. As shown, the collector efficiency and outlet temperature difference for binary nanofluids have the highest value in both volume fractions compared to CuO nanofluids. The outlet temperature differences for C1 and B1 nanofluids are 23.06 K and 26.24 K, respectively. Therefore, by adding 0.04 % of Al_2O_3, the outlet temperature difference increases by 3.18 K. The extinction coefficient of the binary nanofluid, which is a combination of two nanofluids, is approximately equivalent to the total extinction coefficient of its constituent parts. This phenomenon causes more energy absorption. As a result, higher temperature differences are observed for binary nanofluids due to their higher extinction coefficients than CuO nanofluids.

Fig. 11 shows the collector efficiency and outlet temperature difference for EG-based nanofluids. As can be seen, improving the volume fraction of CuO/EG nanofluid boosts the collector efficiency by 1.73 %, while for binary/EG nanofluid, the efficiency enhancement is 0.38 %. It is worth noting that the temperature variation rate and collector efficiency improvement in the EG-based fluid is more significant than in water-based.

By comparing Figs. 10 and 11, it is obvious that the water-based nanofluids are superior to EG-based nanofluids in terms of collector efficiency. This is because the temperature rise in the EG-based nanofluids is more than water-based ones, and due to the higher temperature, there is a more significant heat loss. Also, the heat loss of each case is presented in Fig. 12 for a more detailed analysis. It is evident that the heat loss is higher

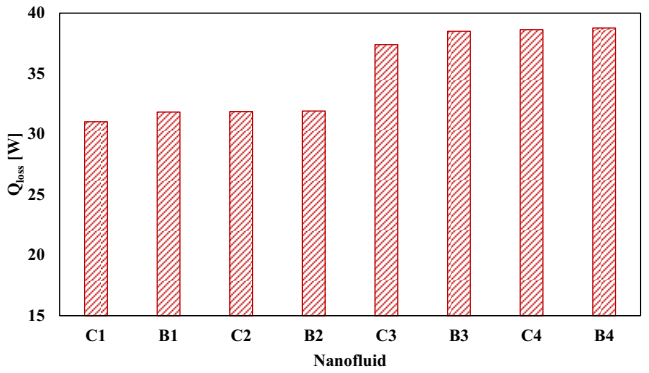


Fig. 12. Heat losses of different working fluids.

using binary nanofluid than both EG and water-based CuO nanofluid. It should be noted that EG or even a mixture of water and EG can be used in a more comprehensive temperature range, because its boiling and freezing temperatures are higher than water. As a result, EG-based nanofluids have a lower collector efficiency compared to water-based ones. Although, EG could be a better base fluid for solar collectors in terms of its wide range of covering temperatures [34].

Fig. 13 (a) illustrates the temperature distribution on the upper surface of the collector for two different volume fractions of the CuO/water and CuO/EG nanofluids. According to the findings, the working fluid temperature is low at the collector inlet and has a higher capacity to absorb energy. Therefore, the temperature rises constantly as it moves along the collector. The maximum temperature for C1, C2, C3, and C4 nanofluids are 316.05 K, 319.55 K, 323.75 K, and 328.17 K, respectively. It is clear that using EG-based nanofluids, the temperature growth between C1 and C3 nanofluids is 2.37 %, while for the C2 and C4 nanofluids is 2.62 %.

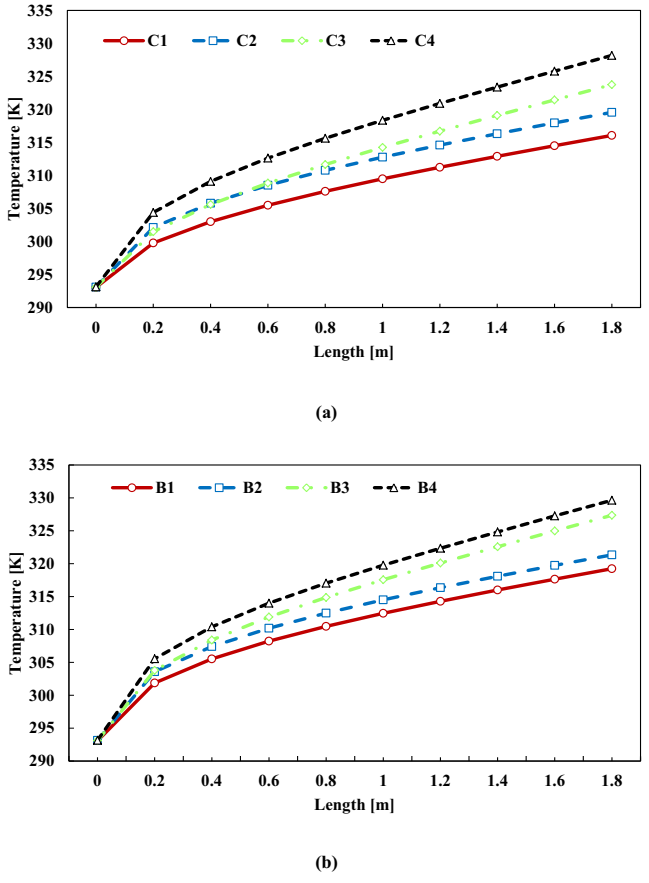


Fig. 13. Temperature distribution on the collector surface using (a) CuO nanofluids and (b) binary nanofluids.

Fig. 13 (b) presents the temperature distribution on the upper surface of the collector for two different volume fractions of the binary/water and binary/EG nanofluids. The surface temperature of the collector exhibits an increasing and uniform trend, which is comparable to the findings presented in Fig. 13 (a). It is obtained that the maximum temperature for B1, B2, B3, and B4 are 319.24 K, 321.34 k, 327.35 K, and 329.64 K, respectively. By comparing the collector surface temperatures using B1 and B3 nanofluids, it can be concluded that the collector surface temperature grows using EG-based nanofluids by 2.47 %. Using B2 and B4 nanofluids, this temperature enhancement equals 2.52 %. Considering the results in Fig. 13 (a) and (b), it is worth noting that by increasing the volume fraction of binary nanofluids, surface temperature variations are less than using CuO nanofluids. This also shows that the extinction coefficient of nanofluid in low volume fraction significantly affects the temperature distribution of the collector surface; however, increasing volume fraction, especially with using binary nanofluid, decreases this effect.

4.2. Effect of inlet velocity

The efficacy of working fluid inlet velocity on collector thermal performance is examined in this section. The simulation performed in the previous sections was considered at a velocity of 0.001 $\frac{m}{s}$. Fig. 14 displays the temperature contour of the collector outlet using B1 nanofluid in the different inlet velocities, which is ranged between 0.001 $\frac{m}{s}$ (Re = 33.64) to 0.003 $\frac{m}{s}$ (Re = 100.91). As shown, enhancement of the inlet velocity improved the Re of the flow, which causes the fluid to have a lower temperature and non-uniform.

Fig. 15 (a) presents the simulation results of two types of nanofluids (C1 and B1) at different inlet velocities. The results reveal that raising the inlet velocity lowers the temperature on the surface of the collector dramatically. Increasing inlet velocity from 0.001 $\frac{m}{s}$ to 0.003 $\frac{m}{s}$ drops the surface temperature from 319.24 K to 308.22 K for the B1 nanofluid, based on findings. Also, the reduction slope is first higher and decreases with increasing velocity. The temperature drop for the C1 nanofluid is estimated to be 10.58 K. Increasing the working fluid velocity decreases the absorption time of solar energy by nanofluids. Therefore, the reduction of temperature occurs at the collector surface.

Fig. 15 (b) depicts the impact of inlet velocity on the collector efficiency and heat losses. It is obvious that the temperature of the surface is more significant at low inlet velocities, which causes further heat to be lost to the environment than at higher inlet velocities. As a result, it can be deduced that the collector efficiency improves as the fluid inlet velocity enhances. According to Fig. 15 (b), the highest thermal efficiencies for C1 and B1 nanofluids occur at the velocity of 0.003 $\frac{m}{s}$, which are 70.51 % and 71.63 %, respectively.

4.3. Effect of MPCM slurry

This section scrutinizes the impact of MPCM slurry on the water-based nanofluid. To simulate the MPCM slurry in this study, two different models are employed:

- Single-phase model
- Two-phase model

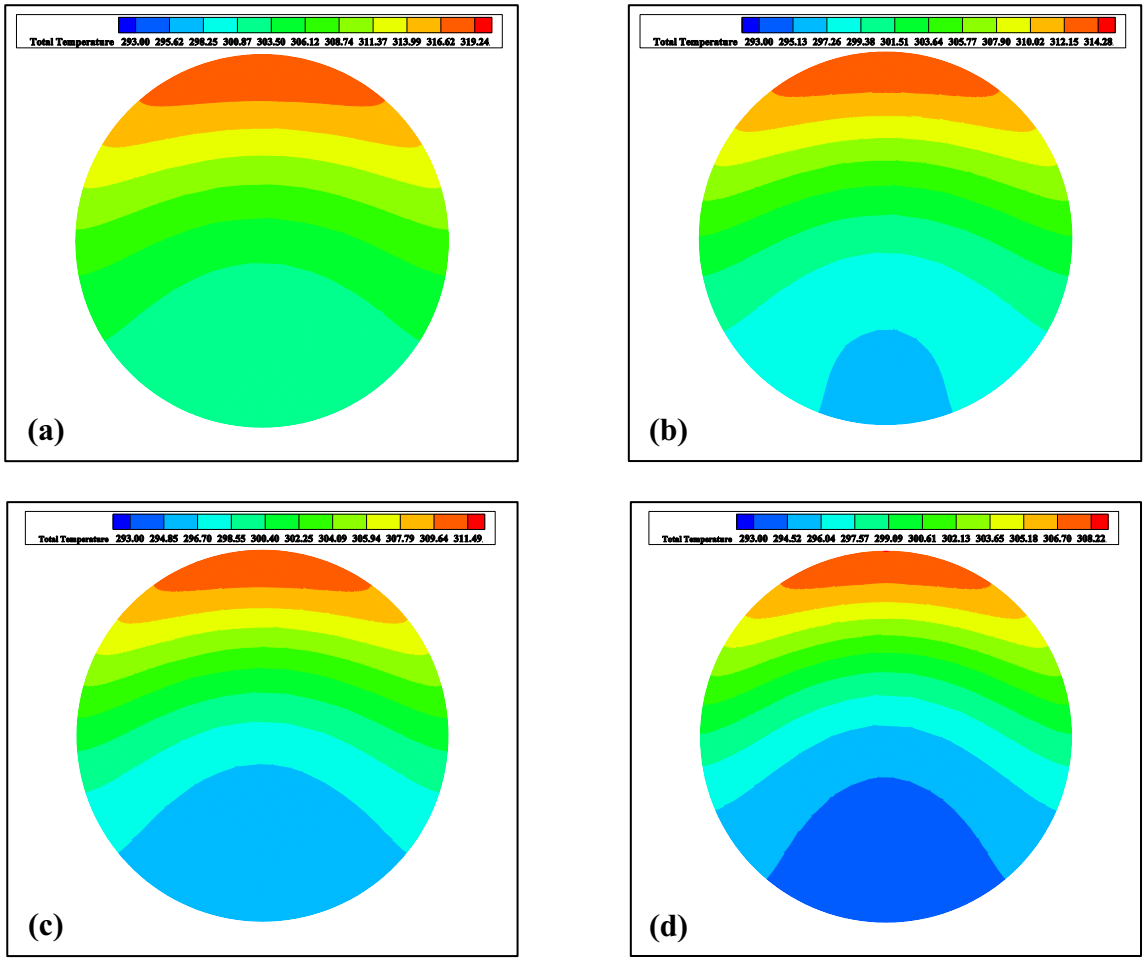


Fig. 14. Temperature contour at the collector outlet using B1 at (a)$\boldsymbol{v} = 0.001\frac{\boldsymbol{m}}{\boldsymbol{s}}$, Re = 33.64 (b) $\boldsymbol{v} = 0.0015\frac{\boldsymbol{m}}{\boldsymbol{s}}$, Re = 50.46 (c)$\boldsymbol{v} = 0.002\frac{\boldsymbol{m}}{\boldsymbol{s}}$, Re = 67.28 (d)) $\boldsymbol{v} = 0.003\frac{\boldsymbol{m}}{\boldsymbol{s}}$, Re = 100.91.

The C2 nanofluid is chosen to compare the influence of the latent heat of MPCM on the efficiency and the temperature distribution at the collector outlet. It should be mentioned that the optical properties are presumed to be constant during phase change. Table 6 shows the properties of the MPCM in the single-phase model by considering the 5 % volume fraction (α_p). Also, it is important to point out that the specific heat of the PCM varies with temperature and increases significantly when the phase change occurs. The outlet temperature contour of the collector is depicted in Fig. 16 (a) by employing the MPCM slurry and running it through the single-phase model. It can be observed that the maximum temperature at the collector outlet is reduced by 0.45 K compared to that without using MPCM. The reason for this phenomenon is that by inserting the MPCM slurry in the collector, the ability of HTF to absorb heat is improved, and the amount of heat transferred from the inner tube to the working fluid improves. Therefore, a drop in temperature occurs at the outlet due to the latent heat contained in the MPCM. Based on Eqs. (5)-(7), the latent heat of the MPCM directly impacts its specific heat capacity and increases during a transition. Thus, the thermal efficiency of the collector elevates by increasing the specific heat capacity.

The properties of the MPCM are applied to the numerical simulation in a separate phase for solving the mixture model. Also, due to the complexity of the continuity, momentum, and energy equations in the two-phase model, the simulation run time is longer compared to the single-phase model. Fig. 16 (b) illustrates the result of the two-phase model. Temperature fluctuations are more pronounced in the two-phase model, which is evident through comparison. The temperature is decreased by 0.2 % (0.63 K) compared to the use of C2 nanofluid in the mixture model.

Fig. 17 shows the efficiency and outlet temperature difference using C2 nanofluid and C2/MPCM slurry. Based on the findings, the collector efficiency utilizing the single-phase and two-phase models is 76.18 % and 76.28 %, respectively, with a relative difference of 0.13 %. The results in both models are very close to each other; therefore, due to the longer computational time of the mixture model, the single-phase model is utilized.

In addition, Fig. 18 illustrates the maximum temperature of the collector surface and heat losses of C2 nanofluid, C2/MPCM slurry single-phase, and mixture models. It is clear that using MPCM slurry not only has a higher efficiency but also has a lower heat loss in comparison to the usage of C2 nanofluid. Moreover, MPCM slurry reduced heat losses by 5.84 % compared to C2 nanofluid. By comparing the simulation results of CuO nanofluids, binary nanofluids, and a combination of CuO nanofluid and MPCM, the collector efficiency using binary nanofluids is lower than that of M1 (MPCM + 0.002 % CuO/water). Consequently, by utilizing the MPCM slurry, the best possible thermal efficiency is obtained. Also, the higher heat capacity of the MPCM than CuO nanofluid contributed to a 4.53 % improvement in the collector thermal efficiency.

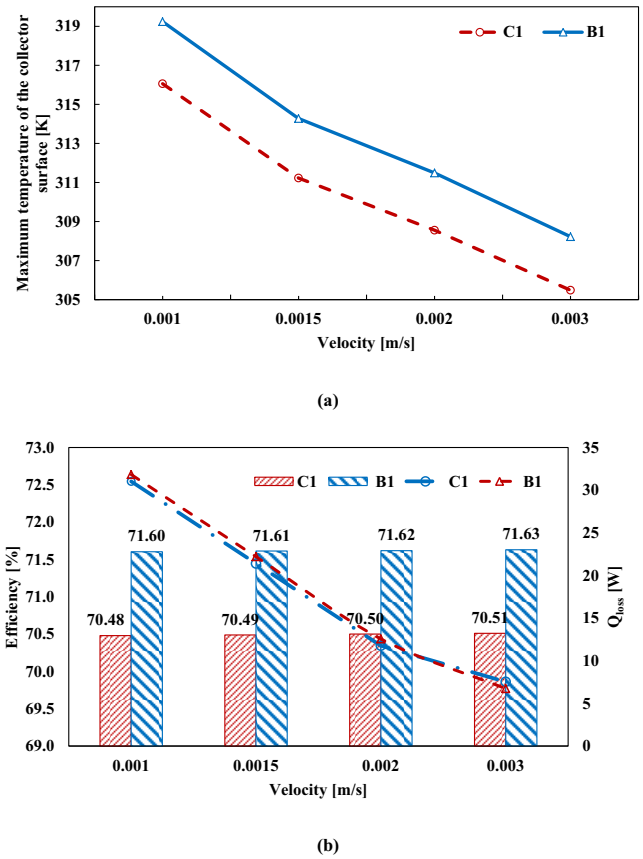


Fig. 15. Effect of inlet velocity on (a) maximum temperature of the collector surface, (b) collector efficiency and heat losses.

5. Conclusion

In this study, the numerical investigation of the DAETC using a combination of nanofluids and MPCM was performed by developing a three-dimensional model using ANSYS-Fluent 2021 R2. Besides, a parametric analysis, including working fluid types, nanofluid volume fractions, and flow velocity, was conducted. The conclusions of the study are listed below:

- The influence of the volume fraction of nanofluid on collector performance revealed that by enhancing the volume fraction, the nanofluid extinction coefficient was improved, which rose the collector outlet temperature dramatically, even in low-volume fractions. On the other hand, outlet temperature is a significant parameter that affects collector efficiency. Improving the volume fraction of CuO/water, CuO/EG, binary/water, and binary/EG nanofluids boosted the collector efficiency by 1.17 %, 1.73 %, 0.12 %, and 0.38 %, respectively.
- By increasing the volume fraction, the increase of outlet temperature using binary nanofluid is less than that using CuO nanofluid in both water and EG-based fluids.
- Although the EG-based nanofluids had a larger outlet temperature than the water-based nanofluids, the latter had higher thermal efficiency. This was because the collector heat loss to the ambient was more using EG as the base fluid than water-based.
- As the inlet velocity of the fluid increases, the collector outlet temperature drops because the time required for the nanofluid to absorb solar energy is reduced. Moreover, heat loss to the ambient declined due to the decrease in collector temperature. Therefore, the collector efficiency is enhanced by improving inlet velocity. The maximum thermal efficiencies for C1 and B1 were 70.51 % and 71.63 %, respectively, at the velocity of 0.003 $\frac{m}{s}$.

Table 6
Effective properties of MPCM with $\alpha_p = 5\%$ [1].

Density ($\frac{kg}{m^3}$)	Specific heat ($\frac{J}{kg.K}$)	$\frac{W}{m.K}$ Melting point (K)	Thermal conductivity ($\frac{W}{m.K}$)	Viscosity ($\frac{kg}{m.s}$)
995	4077	310	0.6	192,660

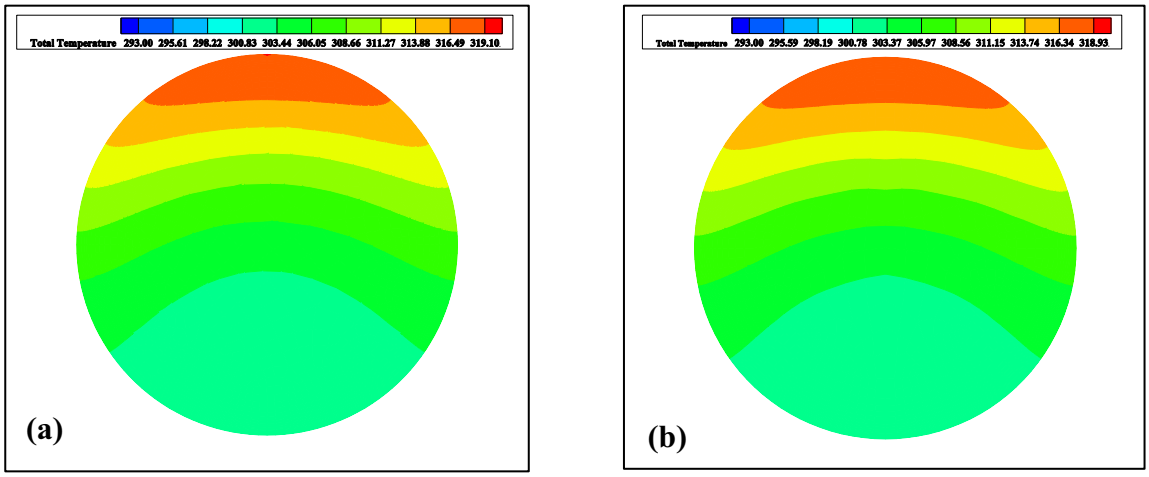


Fig. 16. Outlet temperature contour of the collector using M1 (a) single-phase model (b) two-phase model.

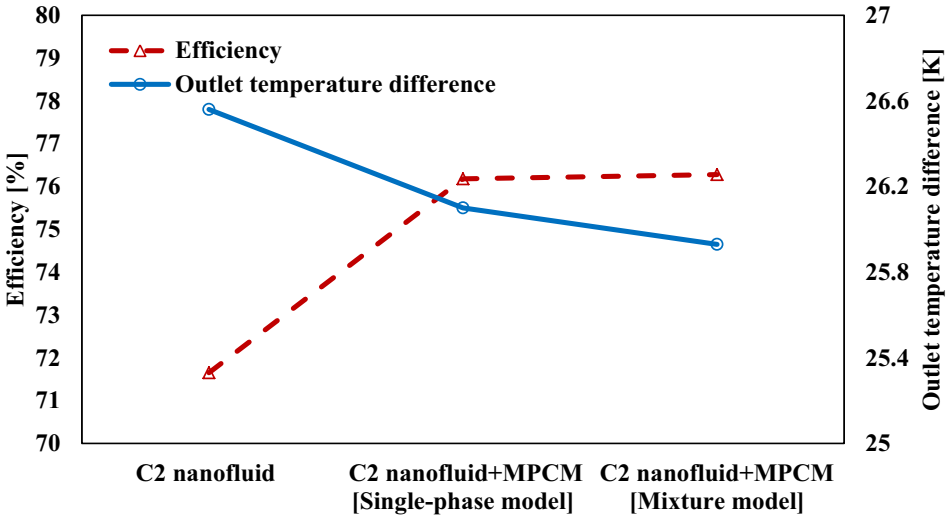


Fig. 17. Efficiency and outlet temperature difference using C2 and C2/MPCM slurry.

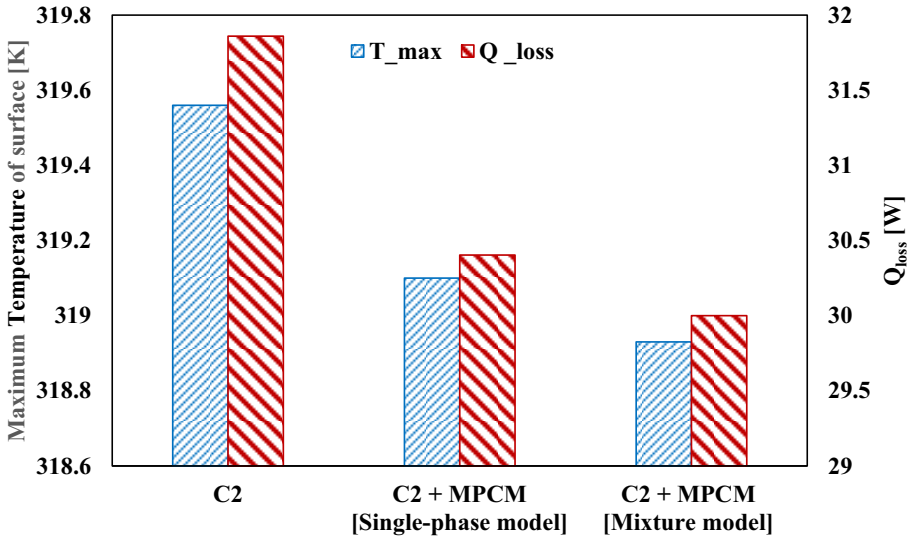


Fig. 18. Maximum temperature of collector surface and heat losses using C2 and C2/MPCM slurry.

- The collector outlet temperature decreased when the MPCM was added to the nanofluid (C2). However, the collector thermal efficiency improved by 4.53 % due to the larger heat capacity of the MPCM. Also, MPCM slurry lowered heat losses by 5.84 % as compared to C2 nanofluid.
- Simulation of the MPCM slurry was accomplished using both single-phase and two-phase (mixture) models, and the results showed that they had an acceptable agreement with each other. The collector efficiency was found to be 76.18 % for the single-phase model and 76.28 % for the mixture model, respectively. The combination of CuO (0.002 %) nanofluid and MPCM produced the best collector efficiency.

Data availability

No data was used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] M. Eisapour, A.H. Eisapour, M. Hosseini, P. Talebizadehsardari, Exergy and energy analysis of wavy tubes photovoltaic-thermal systems using microencapsulated PCM nano-slurry coolant fluid, Appl. Energy 266 (2020).
[2] S. Delfani, M. Karami, M.A. Akhavan Behabadi, Experimental investigation on performance comparison of nanofluidbased direct absorption and flat plate solar collectors, Int. J. Nano Dimens. 7 (1) (2016) 85–96.
[3] S. Ladjevardi, A. Asnaghi, P. Izadkhast, A. Kashani, Applicability of graphite nanofluids in direct solar energy absorption, Sol. Energy 94 (2013) 327–334.
[4] P. Raj, S. Subudhi, A review of studies using nanofluids in flat-plate and direct absorption solar collectors, Renew. Sust. Energ. Rev. 84 (2018) 54–74.
[5] M. Karami, M. Raisee, S. Delfani, Numerical investigation of nanofluid-based solar collectors, IOP Conference Series: Materials Science and Engineering,

Volume 64, 2nd International Conference on Structural Nano Composites (NANOSTRUC 2014) 20–21 May 2014, Madrid, Spain.
[6] C. Guo, C. Liu, S. Jiao, R. Wang, Z. Rao, Introducing optical fiber as internal light source into direct absorption solar collector for enhancing photo-thermal conversion performance of MWCNT-H2O nanofluids, Appl. Therm. Eng. 173 (2020).
[7] M. Esmaeili, M. Karami, S. Delfani, Performance enhancement of a direct absorption solar collector using copper oxide porous foam and nanofluid, Int. J. Energ. Res. 44 (7) (2020) 5527–5544.
[8] B.V. Balakin, O.V. Zhdaneev, A. Kosinska, K.V. Kutsenko, Direct absorption solar collector with magnetic nanofluid: CFD model and parametric analysis, Renew. Energy 136 (2019) 23–32.
[9] M. Karami, S. Delfani, M. Esmaeili, Effect of V-shaped rib roughness on the performance of nanofluid-based direct absorption solar collectors, J. Therm. Anal. Calorim. 138 (2019) 559–572.
[10] M. Siavashi, K. Ghasemi, R. Yousofvand, S. Derakhshan, Computational analysis of SWCNH nanofluid-based direct absorption solar collector with a metal sheet, Sol. Energy 170 (2018) 252–262.
[11] M. Eltaweel, A.A. Abdel-Rehim, A.A. Attia, Energetic and exergetic analysis of a heat pipe evacuated tube solar collector using MWCNT/water nanofluid, Case Stud. Therm. Eng. 22 (2020).
[12] T. Sokhansefat, A. Kasaeian, K. Rahmani, A.H. Heidari, F. Aghakhani, O. Mahian, Thermoeconomic and environmental analysis of solar flat plate and evacuated tube collectors in cold climatic conditions, Renew. Energy 115 (2018) 501–508.
[13] M. Sabiha, R. Saidur, S. Mekhilef, O. Mahian, Progress and latest developments of evacuated tube solar collectors, Renew. Sust. Energ. Rev. 51 (2015) 1038–1054.
[14] A. Kumar, Z. Said, E. Bellos, An up-to-date review on evacuated tube solar collectors, J. Therm. Anal. Calorim. 145 (2021) 2873–2889.
[15] A.K. Hamzat, M.I. Omisanya, A.Z. Sahin, O.R. Oyetunji, N.A. Olaitan, Application of nanofluid in solar energy harvesting devices: A comprehensive review, Energy Convers. Manag. 266 (2022).
[16] M.M. Tafarroj, R. Daneshazarian, A. Kasaeian, CFD modeling and predicting the performance of direct absorption of nanofluids in trough collector, Appl. Therm. Eng. 148 (2019) 256–269.
[17] G. O'Keeffe, S. Mitchell, T. Myers, V. Cregan, Modelling the efficiency of a nanofluid-based direct absorption parabolic trough solar collector, Sol. Energy 159 (2018) 44–54.
[18] A. Kasaeian, R. Daneshazarian, F. Pourfayaz, Comparative study of different nanofluids applied in a trough collector with glass-glass absorber tube, J. Mol. Liq. 234 (2017) 315–323.
[19] A. Menbari, A.A. Alemrajabi, A. Rezaei, Heat transfer analysis and the effect of CuO/Water nanofluid on direct absorption concentrating solar collector, Appl. Therm. Eng. 104 (2016) 176–183.
[20] H.K. Gupta, G.D. Agrawal, J. Mathur, Investigations for effect of Al2O3–H2O nanofluid flow rate on the efficiency of direct absorption solar collector, Case Stud. Therm. Eng. 5 (2015) 70–78.
[21] A. Menbari, A.A. Alemrajabi, Y. Ghayeb, Investigation on the stability, viscosity and extinction coefficient of CuO–Al2O3/Water binary mixture nanofluid, Exp. Therm. Fluid Sci. 74 (2016) 122–129.
[22] A. Kasaeian, R. Daneshazarian, F. Pourfayaz, S. Babaei, M. Sheikhpour, S. Nakhjavani, Evaluation of MWCNT/ethylene glycol nanofluid flow in a parabolic trough collector with glass-glass absorber tube, Int. J. Numerical Methods Heat Fluid Flow 30 (1) (2019) 176–205.
[23] F. Javadi, H. Metselaar, P. Ganesan, Performance improvement of solar thermal systems integrated with phase change materials (PCM), a review, Sol. Energy 206 (2020) 330–352.
[24] F. Yazdanifard, M. Ameri, R.A. Taylor, Numerical modeling of a concentrated photovoltaic/thermal system which utilizes a PCM and nanofluid spectral splitting, Energy Convers. Manag. 215 (2020).
[25] A. Hasan, J. Sarwar, H. Alnoman, S. Abdelbaqi, Yearly energy performance of a photovoltaic-phase change material (PV-PCM) system in hot climate, Sol. Energy 146 (2017) 417–429.
[26] P. Feliński, R. Sekret, Experimental study of evacuated tube collector/storage system containing paraffin as a PCM, Energy 114 (2016) 1063–1072.
[27] X. Ma, Y. Liu, H. Liu, L. Zhang, B. Xu, F. Xiao, Fabrication of novel slurry containing graphene oxide-modified microencapsulated phase change material for direct absorption solar collector, Sol. Energy Mater. Sol. Cells 188 (2018) 73–80.
[28] L. Liu, Y. Jia, Y. Lin, G. Alva, G. Fang, Performance evaluation of a novel solar photovoltaic–thermal collector with dual channel using microencapsulated phase change slurry as cooling fluid, Energy Convers. Manag. 145 (2017) 30–40.
[29] G. Serale, F. Goia, M. Perino, Numerical model and simulation of a solar thermal collector with slurry Phase Change Material (PCM) as the heat transfer fluid, Sol. energy 134 (2016) 429–444.
[30] Z. Qiu, X. Zhao, P. Li, X. Zhang, S. Ali, J. Tan, Theoretical investigation of the energy performance of a novel MPCM (Microencapsulated Phase Change Material) slurry based PV/T module, Energy 87 (2015) 686–698.
[31] Y. Zhang, X. Li, J. Li, C. Ma, L. Guo, X. Meng, Solar-driven phase change microencapsulation with efficient Ti4O7 nanoconverter for latent heat storage, Nano Energy 53 (2018) 579–586.
[32] F. Wang, J. Liu, X. Fang, Z. Zhang, Graphite nanoparticles-dispersed paraffin/water emulsion with enhanced thermal-physical property and photo-thermal performance, Sol. Energy Mater. Sol. Cells 147 (2016) 101–107.
[33] A. Menbari, A.A. Alemrajabi, A. Rezaei, Experimental investigation of thermal performance for direct absorption solar parabolic trough collector (DASPTC) based on binary nanofluids, Exp. Therm. Fluid Sci. 80 (2017) 218–227.
[34] A. Menbari, A.A. Alemrajabi, Y. Ghayeb, Experimental investigation of stability and extinction coefficient of Al2O3–CuO binary nanoparticles dispersed in ethylene glycol–water mixture for low-temperature direct absorption solar collectors, Energy Convers. Manag. 108 (2016) 501–510.
[35] F. Ma, P. Zhang, Performance investigation of the direct absorption solar collector based on phase change slurry, Appl. Therm. Eng. 162 (2019).
[36] Q. Yu, A. Romagnoli, R. Yang, D. Xie, C. Liu, Y. Ding, Y. Li, Numerical study on energy and exergy performances of a microencapsulated phase change material slurry based photovoltaic/thermal module, Energy Convers. Manag. 183 (2019) 708–720.
[37] Z. Ma, P. Zhang, Modeling the heat transfer characteristics of flow melting of phase change material slurries in the circular tubes, Int. J. Heat Mass Transf. 64 (2013) 874–881.
[38] N.S. Roberts, R. Al-Shannaq, J. Kurdi, S.A. Al-Muhtaseb, M.M. Farid, Efficacy of using slurry of metal-coated microencapsulated PCM for cooling in a micro-channel heat exchanger, Appl. Therm. Eng. 122 (2017) 11–18.
[39] P. Charunyakorn, S. Sengupta, S. Roy, Forced convection heat transfer in microencapsulated phase change material slurries: flow in circular ducts, Int. J. Heat Mass Transf. 34 (3) (1991) 819–833.
[40] X. Hu, Y. Zhang, Novel insight and numerical analysis of convective heat transfer enhancement with microencapsulated phase change material slurries: laminar flow in a circular tube with constant heat flux, Int. J. Heat Mass Transf. 45 (15) (2002) 3163–3172.
[41] A. Salari, A. Kazemian, T. Ma, A. Hakkaki-Fard, J. Peng, Nanofluid based photovoltaic thermal systems integrated with phase change materials: numerical simulation and thermodynamic analysis, Energy Convers. Manag. 205 (2020).
[42] K. Chopra, V. Tyagi, A. Pandey, R.K. Sharma, A. Sari, PCM integrated glass in glass tube solar collector for low and medium temperature applications: Thermodynamic & techno-economic approach, Energy 198 (2020).
[43] L. Ma, Z. Lu, J. Zhang, R. Liang, Thermal performance analysis of the glass evacuated tube solar collector with U-tube, Build. Environ. 45 (9) (2010) 1959–1967.
[44] M. Simonetti, F. Restagno, E. Sani, M. Noussan, Numerical investigation of direct absorption solar collectors (DASC), based on carbon-nanohorn nanofluids, for low temperature applications, Sol. Energy 195 (2020) 166–175.
[45] B. Chen, X. Wang, R. Zeng, Y. Zhang, X. Wang, J. Niu, Y. Li, H. Di, An experimental study of convective heat transfer with microencapsulated phase change material suspension: laminar flow in a circular tube under constant heat flux, Exp. Therm. Fluid Sci. 32 (8) (2008) 1638–1646.